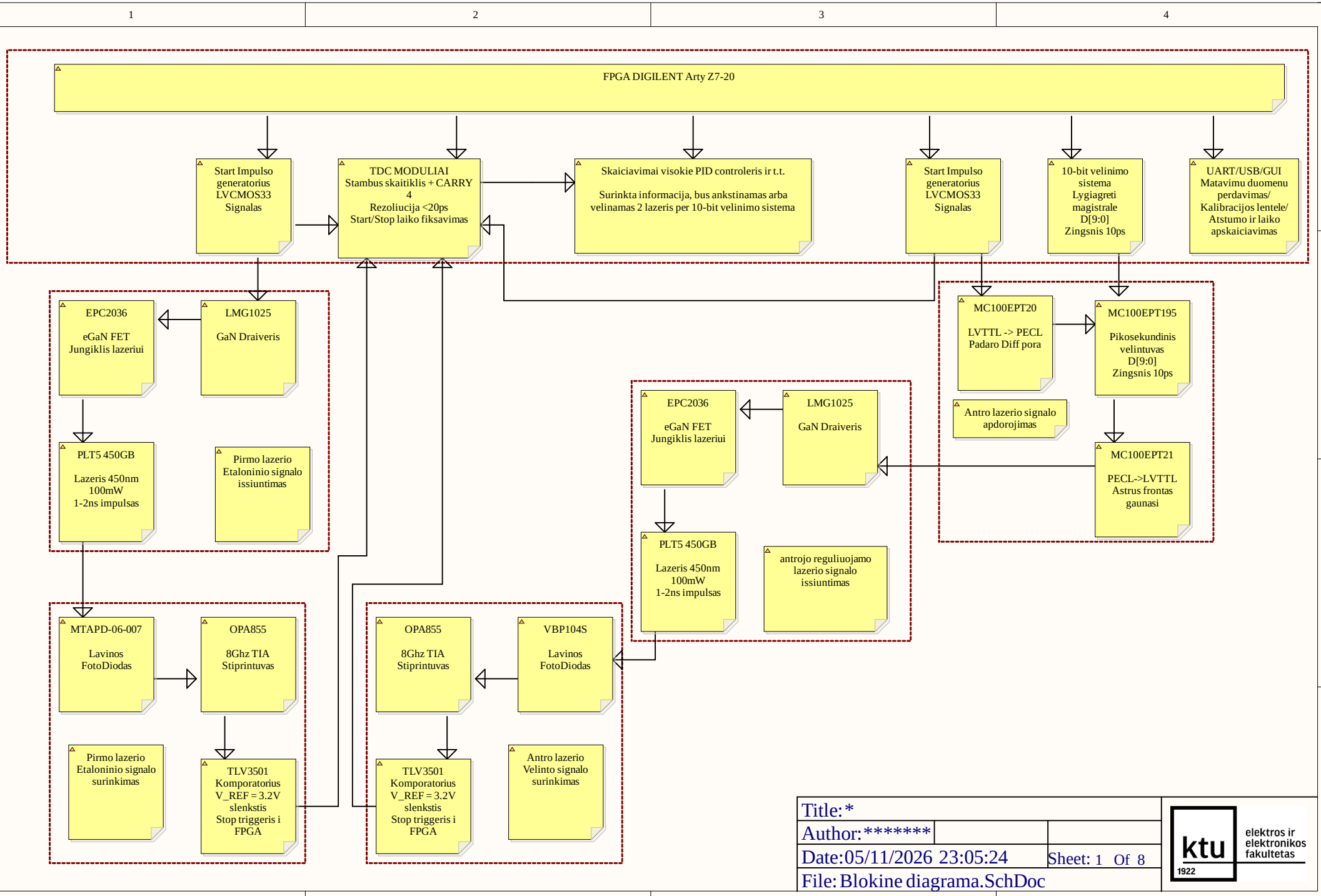

FPGA DIGILENT Arty Z7-20
Start Impulso generatorius LVCMOS33 Signalas
TDC MODULIAI Stambus skaitiklis + CARRY 4 Rezoliucija <20ps Start/Stop laiko fiksavimas
Skaiciavimai visokie PID controleris ir t.t. Surinkta informacija, bus ankstinamas arba velinamas 2 lazeris per 10-bit velinimo sistema
Start Impulso generatorius LVCMOS33 Signalas
10-bit velinimo sistema Lygiagreti magistrale D[9:0] Zingsnis 10ps
UART/USB/GUI Matavimu duomenu perdavimas/ Kalibracijos lentele/ Atstumo ir laiko apskaiciavimas
EPC2036 eGaN FET Jungiklis lazeriui
LMG1025 GaN Draiveris
PLT5 450GB Lazeris 450nm 100mW 1-2ns impulsas
Pirmo lazerio Etaloninio signalo issiuntimas
MC100EPT20 LVTTL -> PECL Padaro Diff pora
MC100EPT195 Pikosekundinis velintuvas D[9:0] Zingsnis 10ps
Antro lazerio signalo apdorojimas
MC100EPT21 PECL->LVTTL Astrus frontas gaunasi
EPC2036 eGaN FET Jungiklis lazeriui
LMG1025 GaN Draiveris
PLT5 450GB Lazeris 450nm 100mW 1-2ns impulsas
antrojo reguliuojamo lazerio signalo issiuntimas
MTAPD-06-007 Lavinos FotoDiodas
OPA855 8Ghz TIA Stiprintuvas
Pirmo lazerio Etaloninio signalo surinkimas
TLV3501 Komporatorius V_REF = 3.2V slenkstis Stop triggeris i FPGA
OPA855 8Ghz TIA Stiprintuvas
VBP104S Lavinos FotoDiodas
TLV3501 Komporatorius V_REF = 3.2V slenkstis Stop triggeris i FPGA
Antro lazerio Velinto signalo surinkimas
Title:*
Author:*******
Date:05/11/2026 23:05:24
Sheet: 1 Of 8
File: Blokine diagrama.SchDoc
ktu 1922
elektros ir elektronikos fakultetas

Matinimas:

Laboratorinis maitinimo saltinis 9V
(Galimas maitinimo saltinio triuksmas, ground loop)

LP38798SD-ADJ/ NOPB
+5V RAIL
OPA855 & TLV3501

LP38798SD-ADJ/ NOPB
+3V3 RAIL
MC100EP195 & SIGNALS

LP38798SD-ADJ/ NOPB
+5V RAIL
GaN Driver

Vcc
9V RAIL
R_charge & C_pulse

LT8331EMSE#PBF
135V Rail
APD Cathode

J103
Vin
1 2 3 4 5 6 7 8 9 10 11 12
Spikelinis conectorius
GND

J103 Spikelinis jungiklis kad sujungt plokste su Lab Bench PSU 9V Vin ir GND

J107 J108 J111 J112
Vin Maitinimai i modulius

J109 J110
135V Input I modulius

J113 J114
135V Output I Modulius

J107 Vin 1 2 Spikelinis conectorius GND
J108 Vin 1 2 Spikelinis conectorius GND
J109 135V 1 2 Spikelinis conectorius GND
J110 135V 1 2 Spikelinis conectorius GND

J111 Vin 1 2 Spikelinis conectorius GND
J112 Vin 1 2 Spikelinis conectorius GND
J113 135V 1 2 Spikelinis conectorius GND
J114 135V 1 2 Spikelinis conectorius GND

Jungtys is FPGA

J101 Reguliuoja MCP Zingsninio velintojo velinima

J102 Jungsis su FPGA per PMOD kadangi signalas gan svarbus, tai tures atskyra GND jungti ant pacio pmod

J104 Jungsis su FPGA per PMOD kadangi signalas gan svarbus, tai tures atskyra GND jungti ant pacio pmod

J101
1 DELAY_D0
2 DELAY_D1
3 DELAY_D2
4 DELAY_D3
5 DELAY_D4
6 DELAY_D5
7 DELAY_D6
8 DELAY_D7
9 DELAY_D8
10 DELAY_D9
11
12
Spikelinis conectorius
GND

J102
1 FPGA_START_TRIG_DELAY
2
Spikelinis conectorius
GND

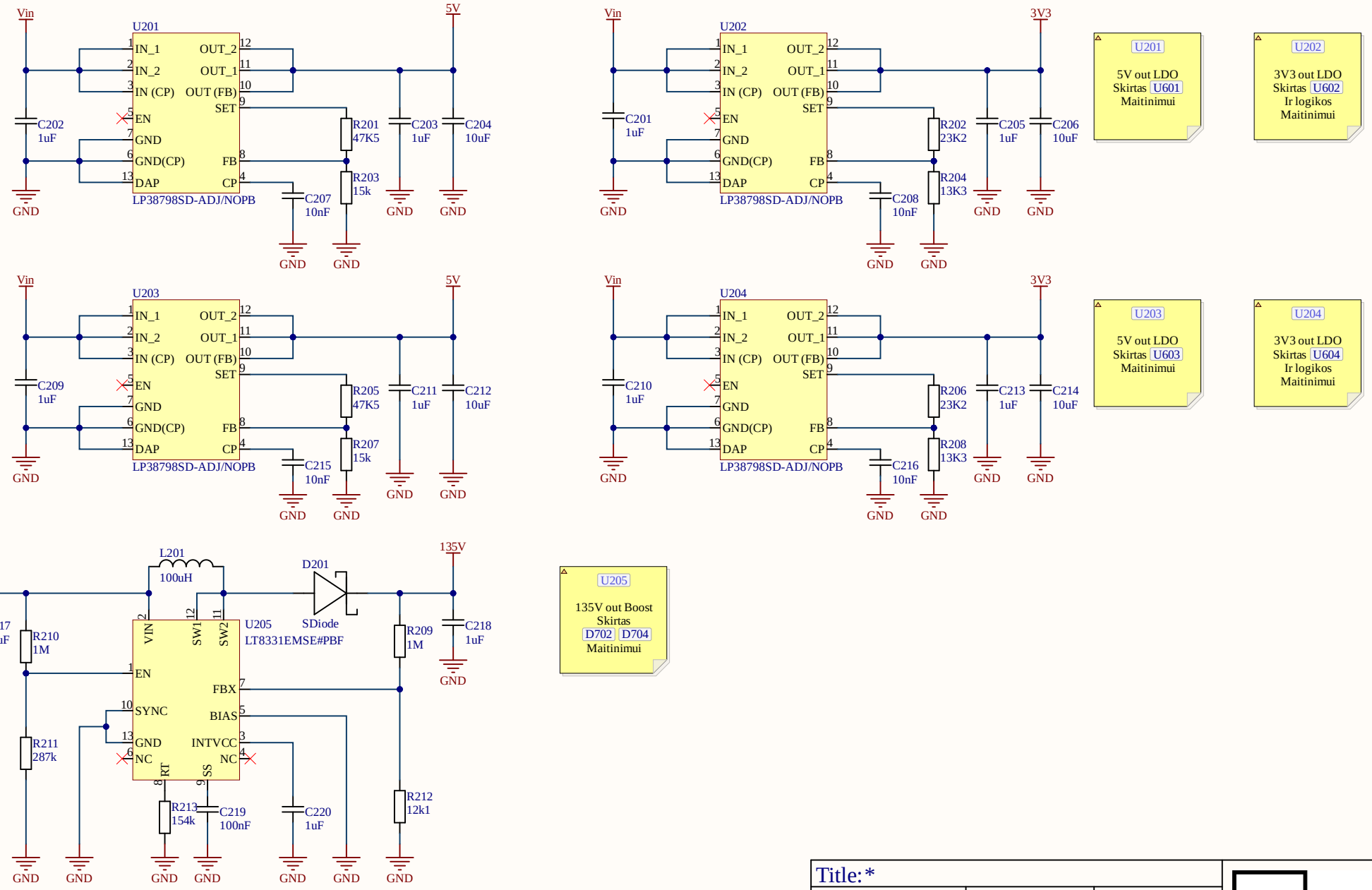
J104
1 FPGA_STOP_PULSE_DELAY
2
Spikelinis conectorius
GND

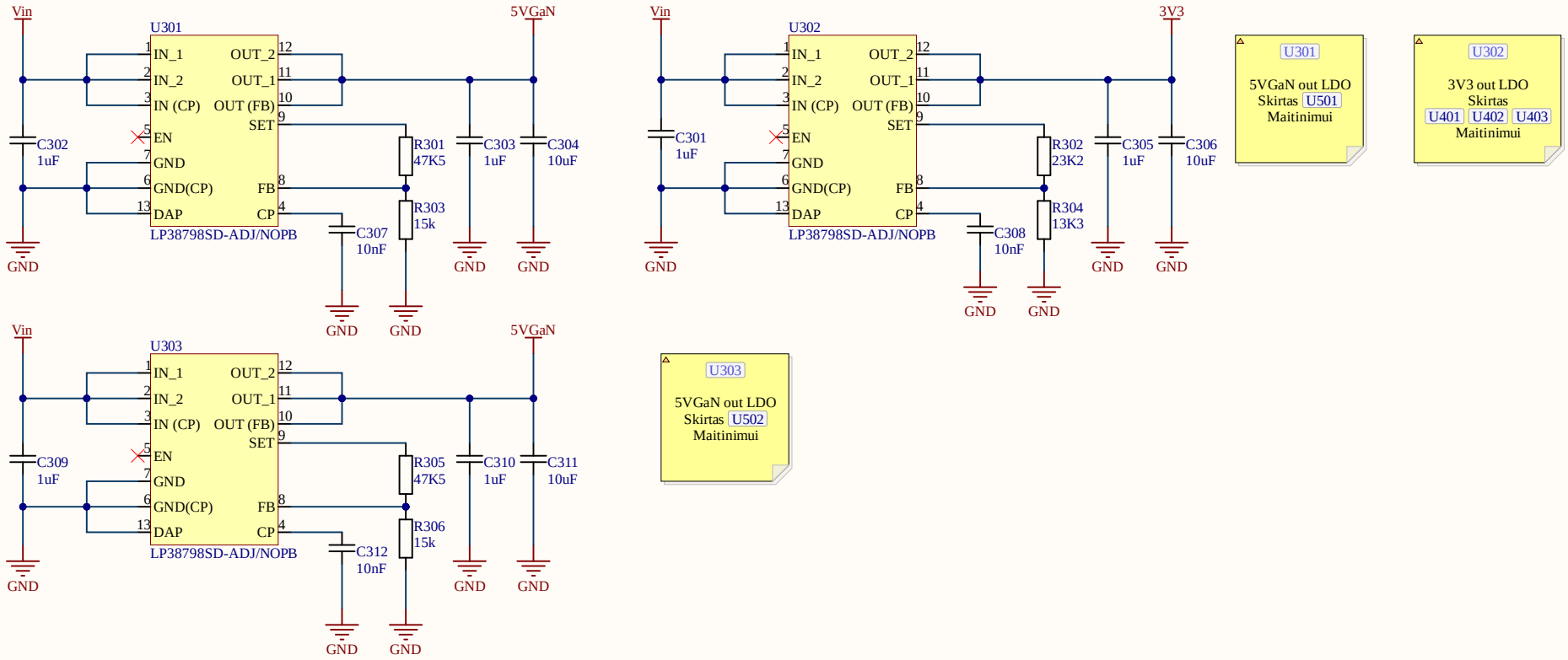
J105
1 FPGA_START_TRIG
2
Spikelinis conectorius
GND

J106
1 FPGA_STOP_PULSE
2
Spikelinis conectorius
GND

Title:*		
Author:*******		
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File: Inputs_Outputs.SchDoc		

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Vin
U201
IN_1
IN_2
IN (CP)
EN
GND
GND(CP)
DAP
OUT_2
OUT_1
OUT (FB)
SET
FB
CP
LP38798SD-ADJ/NOPB
5V
C202
1uF
R201
47K5
R203
15k
C203
1uF
C204
10uF
C207
10nF
U202
3V3
C201
1uF
R202
23K2
R204
13K3
C205
1uF
C206
10uF
C208
10nF
U203
C209
1uF
R205
47K5
R207
15k
C211
1uF
C212
10uF
C215
10nF
U204
C210
1uF
R206
23K2
R208
13K3
C213
1uF
C214
10uF
C216
10nF
U201
5V out LDO
Skirtas U601
Maitinimui
U202
3V3 out LDO
Skirtas U602
Ir logikos
Maitinimui
U203
5V out LDO
Skirtas U603
Maitinimui
U204
3V3 out LDO
Skirtas U604
Ir logikos
Maitinimui
L201
100uH
D201
SDiode
135V
C217
10uF
R210
1M
R209
1M
C218
1uF
U205
LT8331EMSE#PBF
VIN
SW1
SW2
EN
FBX
SYNC
BIAS
INTVCC
NC
RT
SS
R211
287k
R213
154k
C219
100nF
C220
1uF
R212
12k1
U205
135V out Boost
Skirtas
D702 D704
Maitinimui
Title:*
Author:*******
Date:05/11/2026 23:05:24
Sheet: 3 Of 8
File:Rx_PSU.SchDoc
ktu
1922
elektros ir
elektronikos
fakultetas



Title: *			
Author: *******			
Date: 05/11/2026 23:05:24		Sheet: 4 Of 8	
File: Tx_PSU.SchDoc			

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U401 LVCMOS -> PEVL Vertejas

Ateina signalas is FPGA

U401 pakeicia paprasta itampa i difrencialine pora kad suprastu U403

U403 Velintojas

Ateina signalas is U401

U403 Gali velinti nuo 0ps iki 10230ps (10.23ns) su 10ps zingsniu.

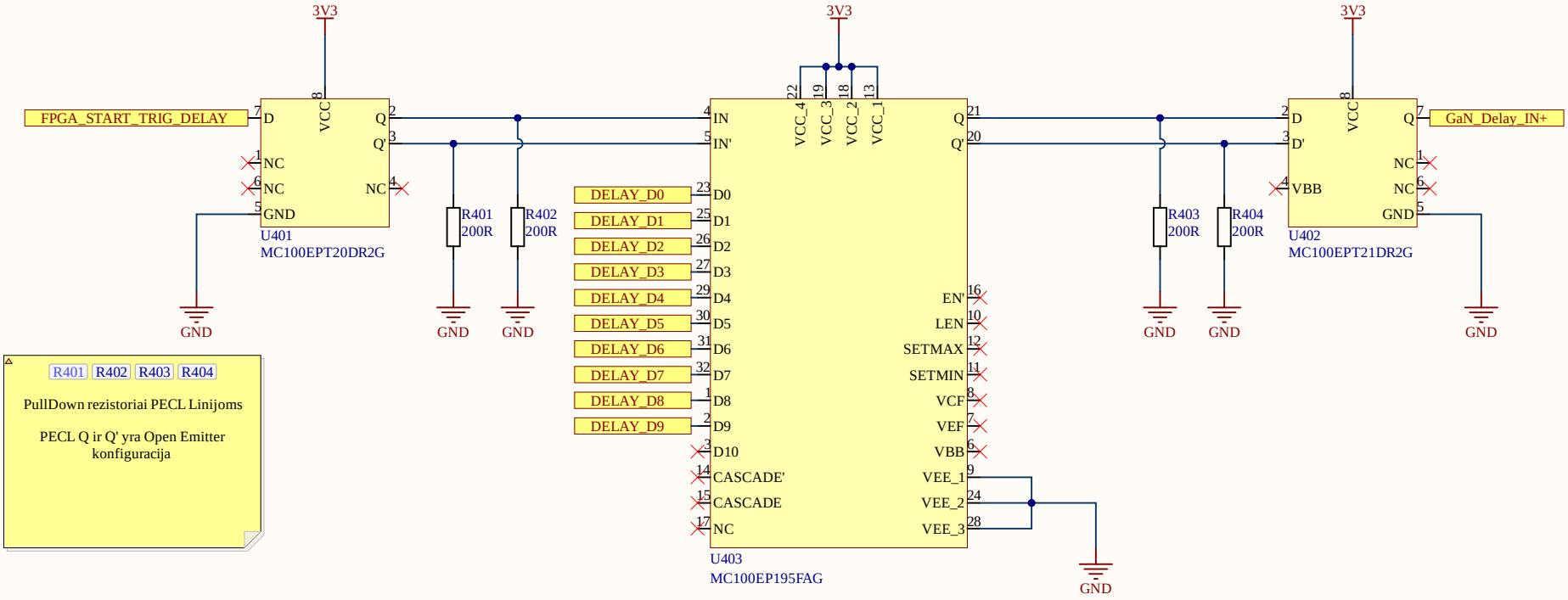
Velinima reguliuoja FPGA per DELAY_D[0:9] 10 bitu magistrale.

Pavelinta signala issiuncia i U402

U402 PEVL -> LVCMOS Vertejas

Ateina diff pora is U403

U402 pakeicia difrencialine pora i paprasta signala kad suprastu U501



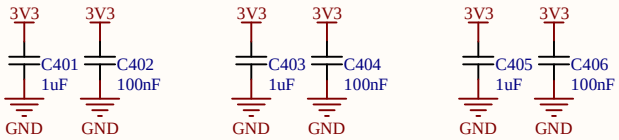
R401 R402 R403 R404

PullDown rezistoriai PECL Linijoms

PECL Q ir Q' yra Open Emitter konfiguracija

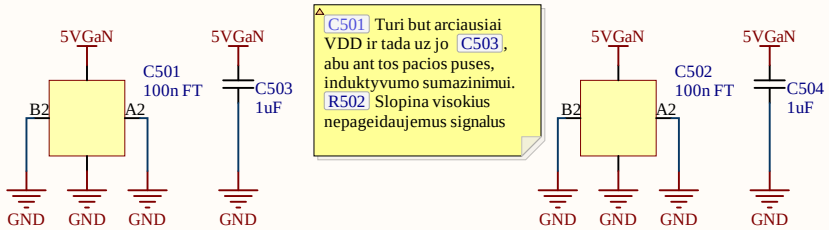
C401 - C406

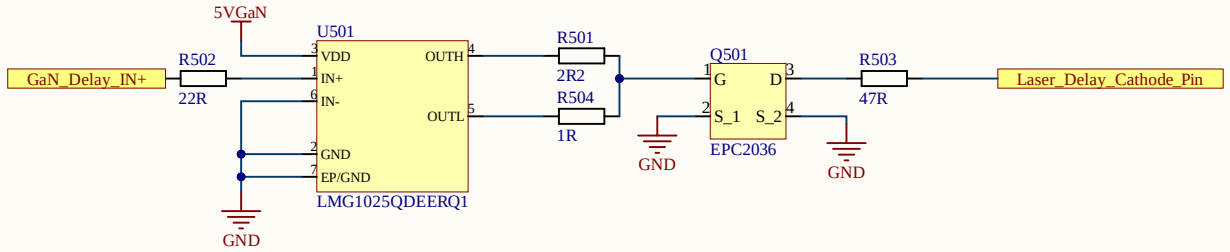
Bus prie U401 - U403 Iejimu

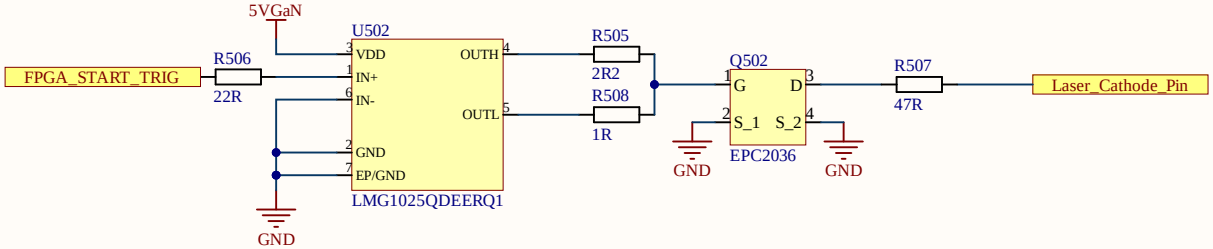


Title:*		
Author:*******		
Date:05/11/2026 23:05:24	Sheet: 5 Of 8	
File: Time_Delay.SchDoc		

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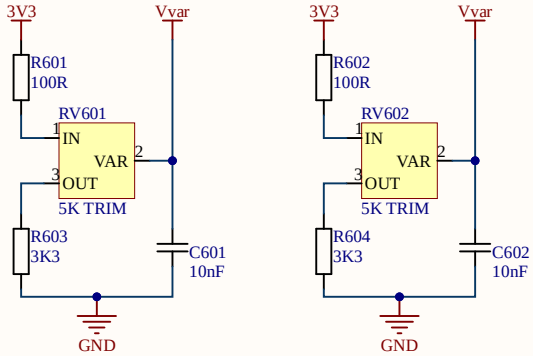


Title:*		
Author:*******		
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File: Laser_Driver.SchDoc		

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C603 - C605 ir C609 - C614

Reikalingi filtravimui, bei stabilios itampos islaikymui.
Kiekviena eile kondensatoriu lituojasi kuo arciau mikruskes
U601 - U604



R602 R604 Apriboja bendra srove teknacia itampos daliklyje
RV602 Leidzia reguliuoti itampa
Nuo 1.30V Iki 3.25V

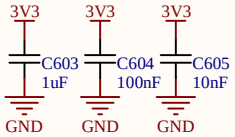
RV602 Reguliuoja U602
RV601 Reguliuoja U604

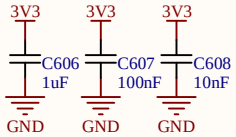
Normali busena:
Fotodiodo srove Ipd = 0mA (tamsu) Vout = 3.3V

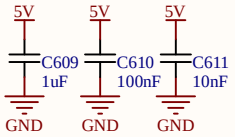
Apsviesta busena:
Fotodiodo srove Ipd = 1mA (blyksnis) Vout = 3.3V - 1mA * 1k = 2.3V
(Dar neskaiciaviau kokias sroves gali paduot fotodiodas, nuo to keisis FB rezistorius)

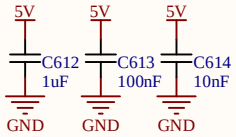
U601 Sustiprina Fotodiodo generuojama signala, kuris pasiekes itampa < Vvar pakeicia U602 komporatoriaus verte i logini 1 (3V3) kuri supranta FPGA.

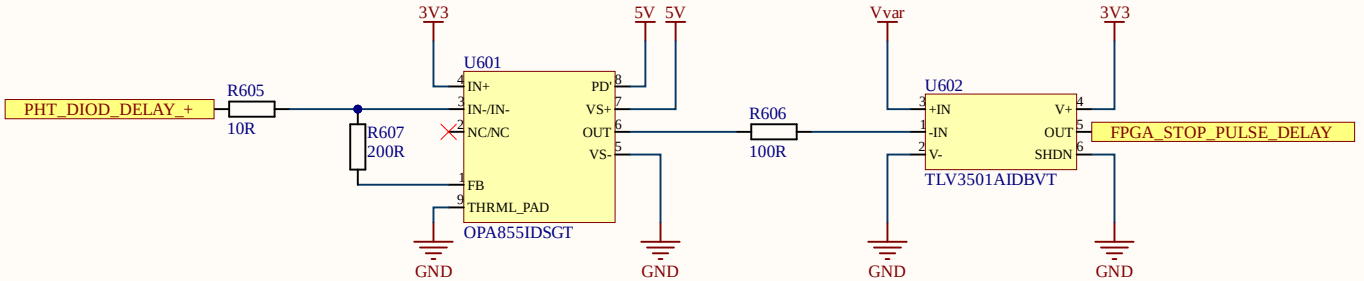
RV602 Reguliuoja Vvar Itampa
absoliucioj tamsoj gal tiks ir 3.25V slenkstis
Apsviestam kambari gali but reikes ir 1.30V slenkscio

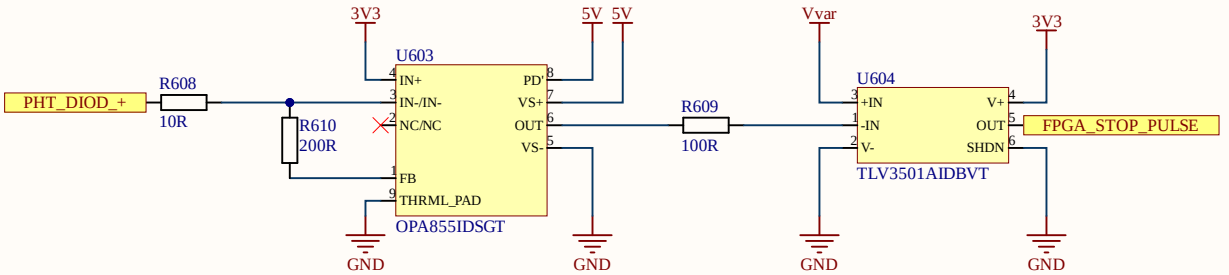












Title:*		
Author:*******		
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File: Light_Detection.SchDoc		

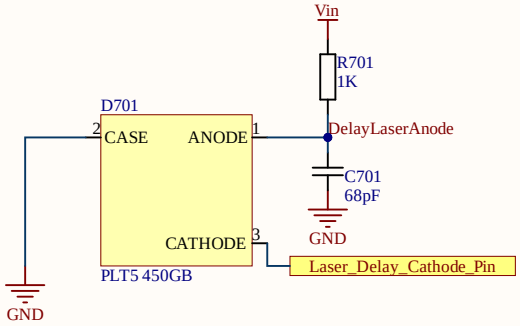
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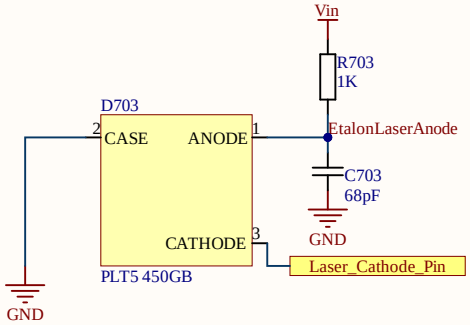
R701 R703 ir C701 C703 Talpinė lazerių iškrova

- R701 / R703 (1k) krauna kondensatorius C701 ir C703 (2nF) iki 9V.

- Atsidarius GaN tranzistoriui, kondensatorius akimirksniu išsikrauna per lazerį (D701 / D703) į žemę ir duoda aštrų sub-ns šviesos impulsą.

- Kondensatoriui išsikrovus lazeris užgęsta pats, o 1k varža apsaugo jį nuo sudegimo.



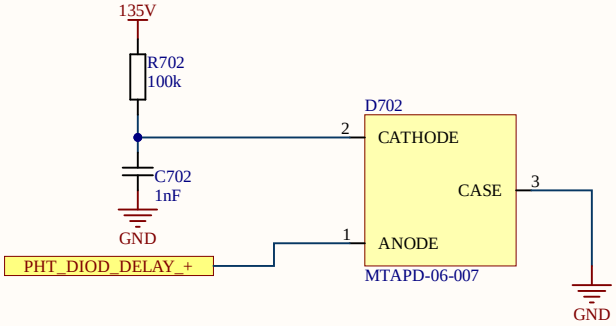


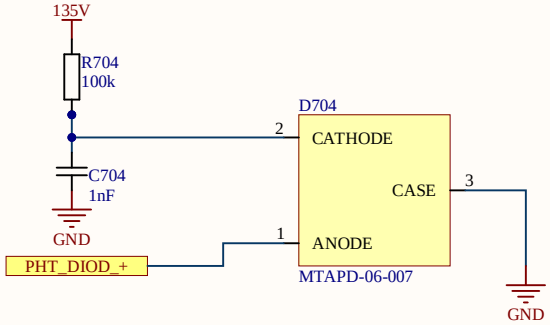
Not found Not found ir Not found Not found Fotodiodų priešįtampis

- Not found / Not found ir Not found / Not found RC filtrai išvalo 5V maitinimo triukšmus.

- Katodai pajungti prie 5V, o Anodai per OPA855 (U601 ir U603) sėdi ties 3.3V (1.7V atbulinis priešįtampis maksimaliai sumažina diodo talpą ir padidina greitį).

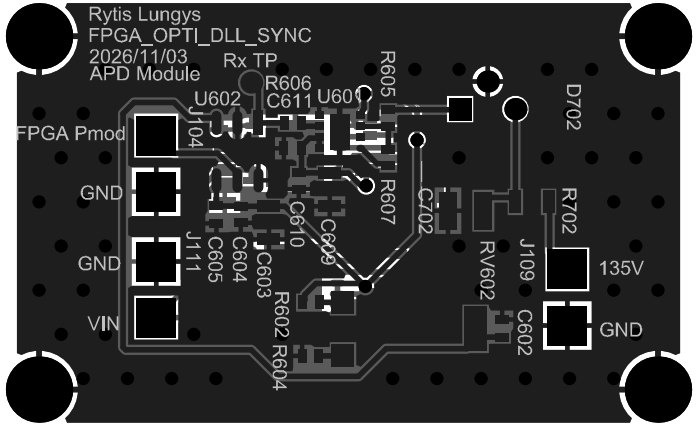
- Šviesos blyksnio srovė siunčiama tiesiai į OPA855 stiprintuvą.

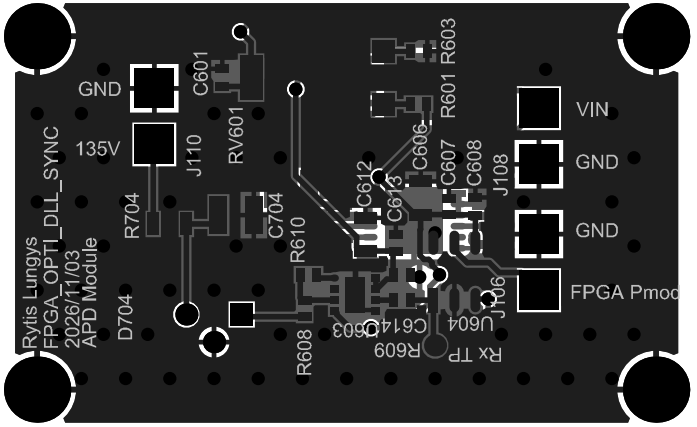


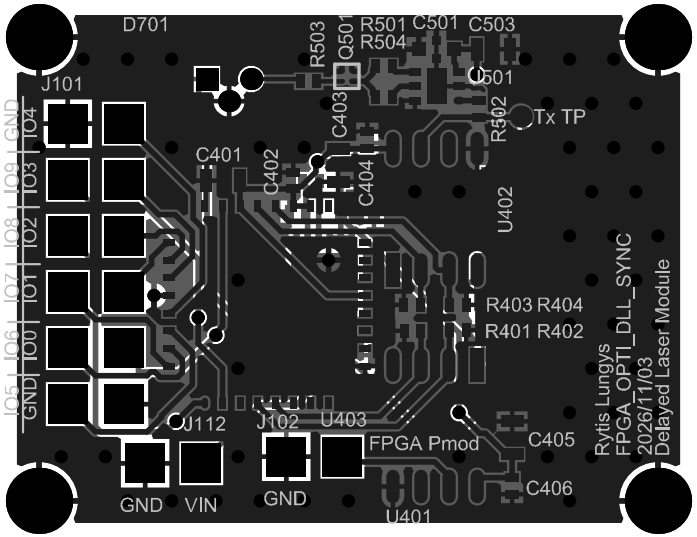


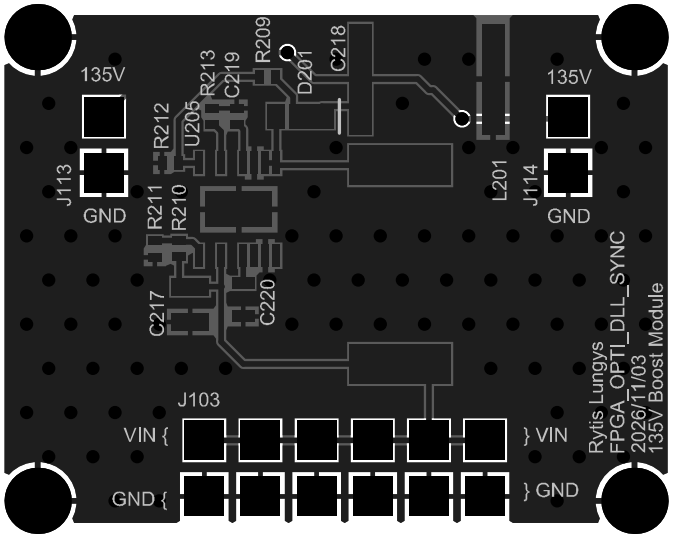
Title:*		
Author:*******		
Date:05/11/2026 23:05:24	Sheet: 8 Of 8	
File: Laser_System.SchDoc		

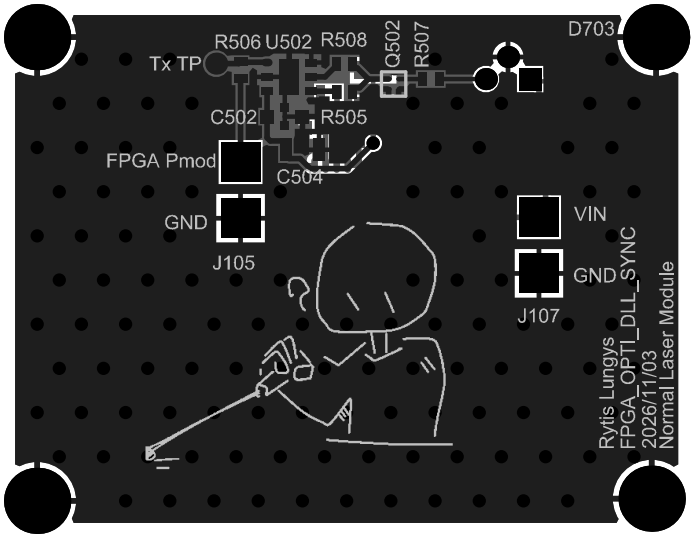
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Rytis Lungys
FPGA_OPTI_DLL_SYNC
2026/11/03
APD Module
Rx TP
R606
C611
U602
U601
R605
D702
FPGA Pmod
J104
GND
GND
VIN
J111
C605
C604
C603
C610
C609
R607
C702
R602
R604
J109
RV602
C602
R702
135V
GND


GND
135V
J110
C601
RV601
R704
C704
R610
D704
R608
R609
Rx TP
U604
C614
C603
C612
C613
C606
C607
C608
R601
R603
J108
J106
VIN
GND
GND
FPGA Pmod
Rytis Lungys
FPGA_OPTI_DLL_SYNC
2026/11/03
APD Module


D701
J101
R503
Q501
R501
R504
C501
C503
U501
C403
R502
Tx TP
C401
C402
C404
U402
R403 R404
R401 R402
GND
IO5
IO6
IO0
IO7
IO1
IO8
IO2
IO9
IO3
GND
IO4
J112
J102
U403
FPGA Pmod
C405
C406
GND
VIN
GND
U401
Rytis Lungys
FPGA_OPTI_DLL_SYNC
2026/11/03
Delayed Laser Module


135V
J113
GND
R212
U205
R213
C219
R209
D201
C218
R211
R210
C217
C220
L201
135V
J114
GND
J103
VIN {
} VIN
GND {
} GND
Rytis Lungys
FPGA_OPTI_DLL_SYNC
2026/11/03
135V Boost Module


D703
R506 U502
R508
Q502
R507
Tx TP
C502
R505
FPGA Pmod
C504
GND
J105
VIN
GND
J107
Rytis Lungys
FPGA_OPTI_DLL_SYNC
2026/11/03
Normal Laser Module